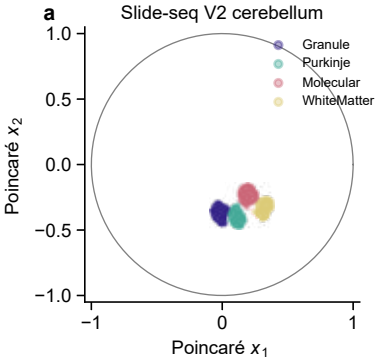
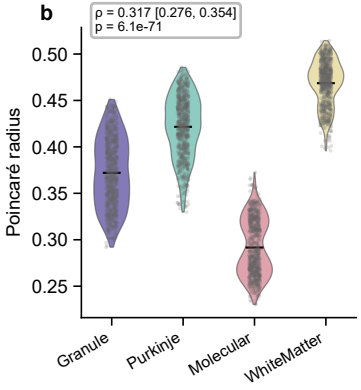
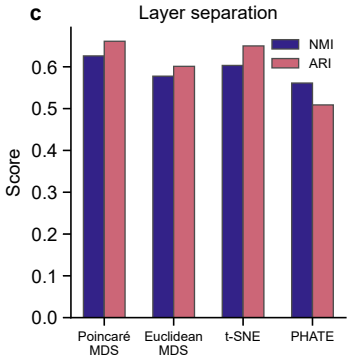
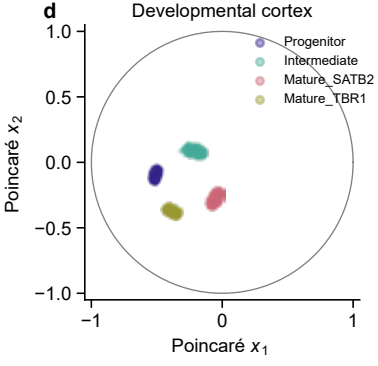
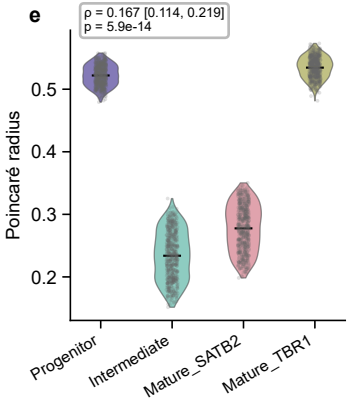
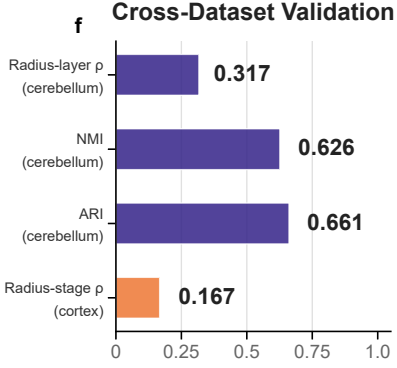

a
Slide-seq V2 cerebellum
Granule
Purkinje
Molecular
WhiteMatter
Poincaré x_2
Poincaré x_1
1.0
0.5
0.0
−0.5
−1.0
−1
0
1


b
ρ = 0.317 [0.276, 0.354]
p = 6.1e-71
Poincaré radius
0.50
0.45
0.40
0.35
0.30
0.25
Granule
Purkinje
Molecular
WhiteMatter


c
Layer separation
NMI
ARI
Score
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Poincaré MDS
Euclidean MDS
t-SNE
PHATE


d
Developmental cortex
Progenitor
Intermediate
Mature_SATB2
Mature_TBR1
Poincaré x_2
Poincaré x_1
1.0
0.5
0.0
−0.5
−1.0
−1
0
1


e
ρ = 0.167 [0.114, 0.219]
p = 5.9e-14
Poincaré radius
0.5
0.4
0.3
0.2
Progenitor
Intermediate
Mature_SATB2
Mature_TBR1


f
Cross-Dataset Validation
Radius-layer ρ (cerebellum)
0.317
NMI (cerebellum)
0.626
ARI (cerebellum)
0.661
Radius-stage ρ (cortex)
0.167
0
0.25
0.5
0.75
1.0